LGAD-W: A Robust and Imperceptible Image Watermarking Scheme Using LBP-Guided Adaptive DCT with Differential Embedding

[1]*Anubhav Rathore, Undergrad, ECE, IIITN*
[2]*Dhananjay Agrahari, Undergrad, ECE, IIITN*
[3]*Sandesh Jat, Undergrad, ECE, IIITN*
[4]*Aman Kanaujiya, Undergrad, ECE, IIITN*

Abstract: Digital images are always susceptible to digital theft, hacking, loss of ownership and many such malpractices. Watermarking becomes a very crucial step against digital malpractices. There exists different methodologies and techniques of embedding watermarks on images. But, balancing imperceptibility and robustness remains a fundamental challenge in blind image watermarking. This paper presents LGAD-W (Local Binary Pattern Guided Adaptive Discrete Cosine Transform Watermarking), a novel integration of established techniques, that achieves high visual quality (PSNR > 40dB) and strong resilience to common digital images attacks. Unlike conventional watermarking approaches that rely on a single feature for block selection, LGAD-W exhibits a two-factor union of block variance and LBP (Local Binary Pattern) texture complexity to identify 8x8 DCT blocks for embedding. Each watermark bit is embedded using differential modulation of mid-frequency DCT coefficients and Arnold scrambling for enhanced security. Extensive experiments on the standard cameraman image have been performed to check the robustness of methodology, under which high Normalised Correlation (NC) of the original watermark and extracted watermark has been achieved.

Keywords: Digital watermarking, DCT, Local Binary Patterns (LBP), differential embedding, robustness, imperceptibility, Arnold transform.

I. INTRODUCTION

The growth of digital multimedia and network distribution has made copyright infringement and unauthorized malpractices on digital data increasingly prevalent. Digital images are very much prone to loss of ownership data, without effective actions. Digital watermarking addresses these concerns by embedding imperceptible, yet recoverable, information directly into the host image. A successful watermarking technique needs to satisfy two essential features: imperceptibility (watermark should not degrade the original image visual quality) and robustness (the watermark must survive common digital processing operations and malicious attacks). Achieving this trade off, along with the possibility of being recoverable, is particularly challenging in blind watermarking, where extraction of watermark does not rely on the original host image.

Transform domain methods, especially Discrete Cosine Transform (DCT) and Discrete Wavelet Transforms (DWT), have proven effective for blind watermarking due to their efficient properties like, energy compaction and compatibility with JPEG compression standards. A common strategy is selecting specific 8×8 DCT blocks of image based on a perceptual metric, such as block variance or texture complexity, and then embedding the watermark by modifying the selected coefficients.

However, relying on a single feature often leads to suboptimal block selection: variance favors high-energy smooth regions but may miss fine textures, while texture measures (e.g., entropy, edge density) can overlook blocks with uniform high

luminance variation. This limitation motivates the use of a union of complementary features to enlarge the pool of suitable embedding locations without sacrificing invisibility.

Local Binary Pattern (LBP) offers a computationally efficient texture description that captures micro-patterns through local neighbourhood comparisons. Although LBP is widely used for texture classification, face recognition and other such applications, LBP has seen limited applications in watermarking block selection as well. Furthermore, most DCT-based schemes embed bits by directly settling coefficients to a fixed value (eg. +- alpha), which introduces a significant amount of distortion when alpha becomes large enough for robustness, or uses quantization index modulation (QIM), that may be sensitive to amplitude scaling.

To overcome these limitations, this paper proposes LGAD-W (LPB-Guided Adaptive DCT Watermarking), a new integration of:

1. Two-factor block selection – the union of top-percentile block variance and top-percentile LBP complexity, ensures a rich set of eligible blocks that are both perceptually significant and texture-rich.

2. Differential DCT embedding – a blind method where a watermark bit is encoded in the difference between two mid-frequency DCT coefficients. This approach reduces distortion because both coefficients are adjusted symmetrically, and extraction reduces to a simple sign test (no threshold tuning).

3. Arnold scrambling – applied to the binary watermark before embedding to enhance security and distribute error patterns.

The remainder of this paper is organized as follows. Section 2 reviews related work on DCT watermarking, block selection strategies, LBP, and differential embedding. Section 3 explains the proposed LGAD-W algorithm, including watermark generation, two-factor block selection, embedding, and extraction. Section 4 discusses the experimental setup and section 5 presents the results, including imperceptibility, robustness under various attacks, and comparisons with baseline methods. Section 6 concludes the paper and outlines future research directions.

II. RELATED WORK

Over the past two decades, the watermarking literature has grown substantially, and have produced a wide range of embedding strategies, each achieved with its own strengths and weaknesses, Our work has been built upon four main pillars of existing research: DCT-based watermarking, block selection methods, differential embedding techniques, and scrambling mechanisms. This section reviews the relevant developments in each area and identifies the gaps that LGAD-W aims to fill.

A. DCT-Domain Watermarking

The discrete cosine transform (DCT) has been widely used in watermarking due to its properties like energy compaction and compatibility with JPEG compression [1], [2]. Early spread-spectrum methods used to embed watermarks in perceptually significant DCT coefficients for being able to survive common signal processing operations[3]. Recent works have optimized the coefficient selection using gradient-based approaches [4] or inter-block correlations with psychovisual thresholds [5]. However, many schemes still suffer from either low PSNR or insufficient robustness under multiple or combined attacks.

B. Block Selection Strategies

Choosing the right 8×8 DCT blocks is crucial. Watermarking in smooth regions leads to visible artifacts, while textured or high-variance regions can tolerate larger modifications [6]. Block variance is a common selection criterion, but it has shown to miss fine-textured blocks with low luminance variations. Researchers have therefore explored complementary texture descriptors, including Local Binary Patterns [7], [8].

C. Local Binary Patterns (LBP) for Watermarking

LBP, first introduced by Ojala et al. [9], is a computationally efficient texture descriptor that compares each pixel with its neighbours to generate a binary code. Although LBP has seen to be used in fragile watermarking [10] and as a pre-selection tool [11], most existing schemes rely on LBP alone. This can be suboptimal because some high-texture blocks have low embedding capacity, while some smooth but high-variance blocks are ignored.

D. Differential Embedding

Differential embedding encodes a watermark bit in the difference between two coefficients instead of modifying a single coefficient directly. This approach is blind, and thus, reduces per-coefficient distortion, simplifying extraction to a sign test [12]. Several works have applied differential principles in hybrid DWT-DCT frameworks [13], [14], reporting robustness against JPEG, noise, and filtering. However, a pure DCT implementation with mid-frequency coefficient pairs is less common.

E. Arnold Scrambling

Arnold (cat) map is an invertible scrambling technique that rearranges pixels in a visually chaotic manner [15]. Applying this method to the watermark before embedding distributes error patterns, adding additional security (the number of iterations acts as a key) [16], [17]. It has been successfully combined with DCT-based watermarking to resist cropping and localized attacks.

III. PROPOSED METHOD

This section describes the LGAD-W watermarking scheme in detail. The overall framework consists of three main phases:

1. Pre-processing (watermark generation and scrambling)
2. Embedding
3. Extraction

Figure 1 shows the block diagram of the embedding process and figure 2 describes the extraction process.

A. Overview

LGAD-W operates on 8×8 DCT blocks of a grayscale host image. It selects eligible blocks using a two-factor union of block variance and LBP texture complexity. Each watermark bit is then embedded by modifying the difference between two mid-frequency DCT coefficients in the selected block. The Extraction is blind that is, it does not require the original image, and uses only the sign of the same difference.

B. Watermark Generation and Scrambling

This paper has implemented a robust watermark generation, as well for the host image. In case of desiring to use your own watermark, this section can be skipped, requiring minute changes in the implementation for ease of flow.

Here, the binary watermark W_orig of size M × M is created as a combination of a border and two diagonals (that is, an "X"). The border thickness is set to M/12 (rounded). The pattern is then inverted so that the background is black (0) and the generated shape is white (1). This pattern is easily recognizable even after moderate attacks. Figure 3 shows the generated watermark.

fig 1. Watermark Embedding Pipeline

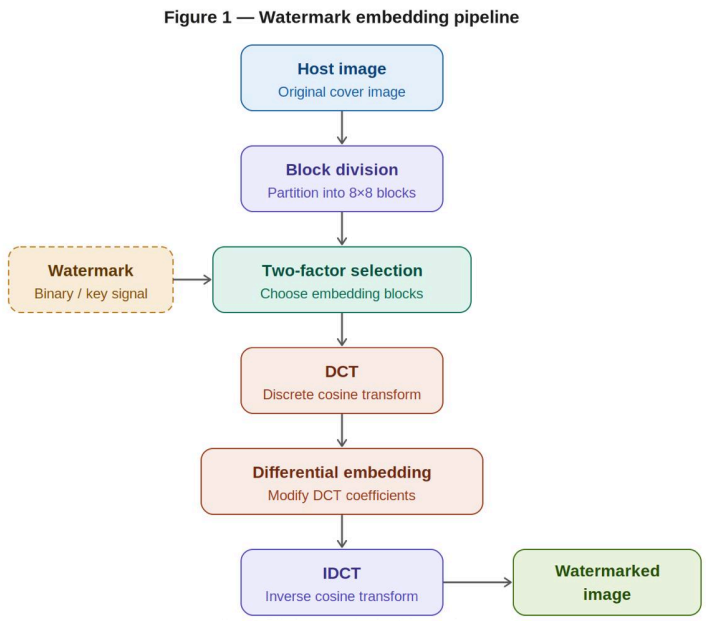
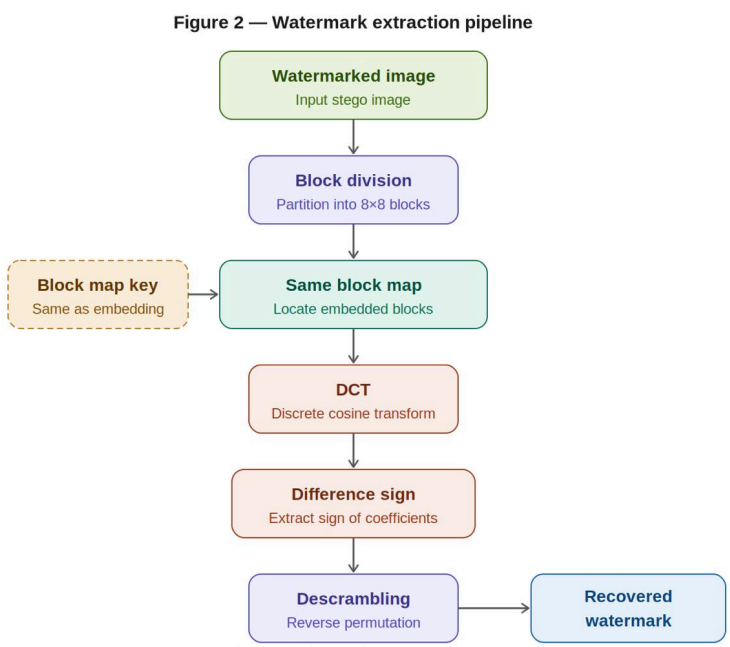


fig 2. Watermark Extraction Pipeline



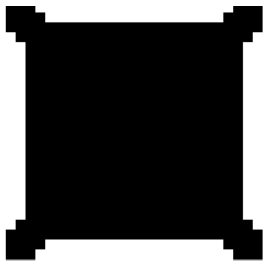


fig 3. Generated Watermark (for experimental purposes). Users can use their own watermarks as well.

Once the watermark is successfully loaded, to disperse error patterns and add security, the watermark is scrambled using Arnold's cat map.
The transform for a pixel at *(i,j)* in a square of size M is given by:
[i_new, j_new] = [(i + j) mod M , (i + 2j) mod M]
And the inverse transform is:
[i, j] = [(2i_new - j_new) mod M , (-i_new + j_new) mod M]

Scrambling is applied for a number of iterations (default = 5). The number of iterations serves as a secret key. After scrambling, the watermark becomes visually unrecognizable. Figure 4 below shows the scrambled watermark.

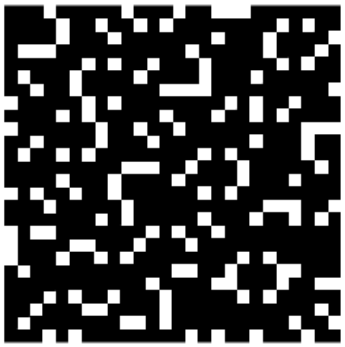


fig 4. Watermark scrambled using Arnold Scrambling

C. Two-Factor Block Selection

This step identifies which 8×8 DCT blocks will carry watermark bits. The selection uses a union of two criteria: block variance and LBP complexity. This ensures that both high-energy smooth blocks and fine-textured blocks are included. The algorithm is as follows and figure 5 shows results.

1. *Divide image into 8x8 blocks*
2. *Compute block variance for each block*
3. *Compute LBP Complexity*
4. *Sum the transition counts over all interior pixels. Higher sum indicates more texture*
5. *Apply percentile thresholds*
6. *Form union map*

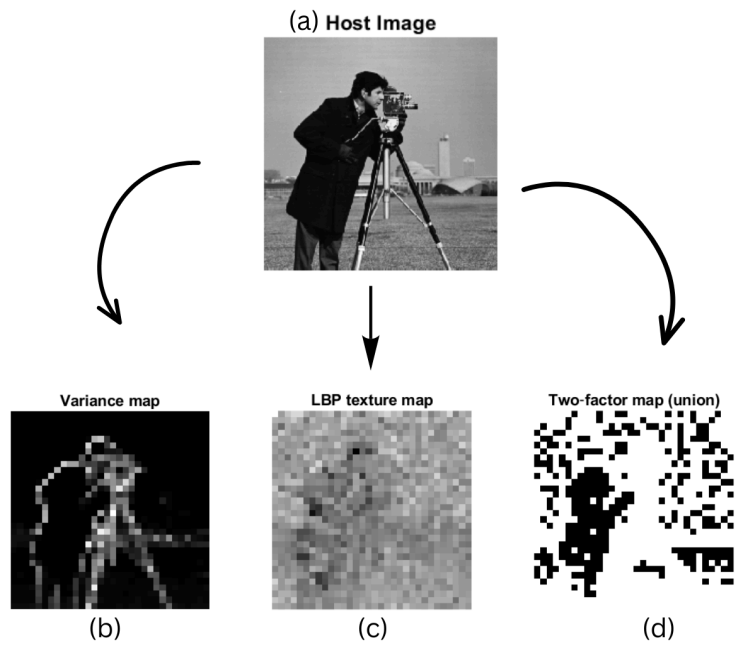


fig 5. (a) Original host image to be watermarked (b) Variance Map © LBP texture map (d) Two-factor union map

D. Differential DCT Embedding

For each eligible block in raster order, one watermark bit is embedded using differential modulation of two mid-frequency DCT coefficients. A set of 16 mid-frequency positions (indices in 8×8 DCT) is pre-defined. The pairs are cycled through blocks sequentially. Using multiple pairs improves robustness and distributes embedding energy.
The algorithm is as follows for each block:

1. *Extract the block and apply 2D DCT*
2. *Select coefficient pair based on k*
3. *Compute current difference*

4. *Target difference for bit b (0 or 1)*
5. *Compute adjustment per coefficient*
6. *Modify both coefficients*
7. *Inverse DCT and clip to valid range [0,1]*
8. *Replace block in watermarked image.*

E. Extraction Process

Extraction follows the same block selection and DCT steps but uses the sign of difference to recover the bits. The receiver must know these same parameters: *variance/LBP Percentiles, Arnold Iterations (number), Coefficient Pair Ordering.*

Then the following algorithm is used to extract the watermark:

1. *Regenerate block map (or use transmitted map)*
2. *For each eligible block:*
    a. *Extract block and compute DCT*
    b. *Select the same coefficient pair (using the same cyclic index)*
    c. *Compute difference: d = Y(p1) - Y(p2)*
    d. *Recover bit: if d > 0 then bit = 1 else bit = 0*
    e. *Store bit in a vector*
    f. *Reshape bits into the M×M matrix. This gives us scrambled watermark*
    g. *Apply inverse Arnold transform with the same number of iterations to obtain extracted watermark.*

Table 1 below discusses a few parameters. And figure 6 shows watermark and extracted watermark under no attack..

Parameter	Typical Value	Role
Embedding strength	0.15	Controls robustness vs. distortion. Higher = more robust but lower PSNR
Variance percentile	40	Top % of blocks by variance selected.
LBP percentile	40	Top % of blocks by LBP complexity selected.
Arnold iterations	5	Security key; distributes error.
Watermark size	Auto-computed	Derived from eligible block count.

Table 1. Method Parameters

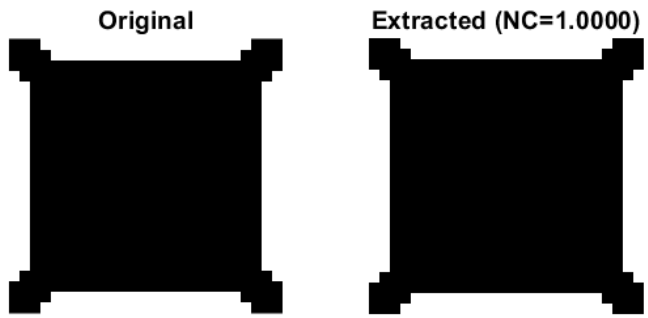


fig 6. Original watermark and extracted watermark. Normalized Correlation of 1 has been achieved, that is, the extracted watermark resembles the original completely.

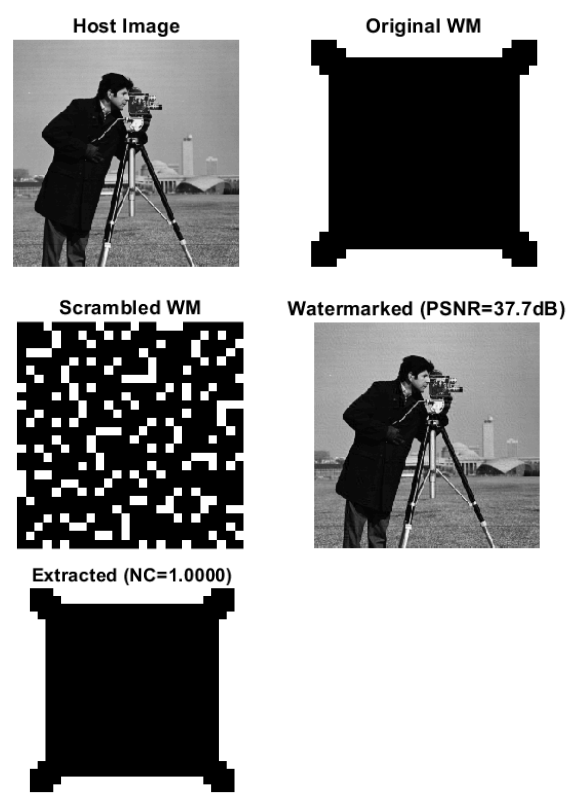
IV. EXPERIMENTAL SETUP and Results

This section describes the test environment and different signal processing operations and attacks done on watermarked images, and then the extracted watermark is compared to the original one, giving us a Normalized Coefficient.

Host Image: The standard grayscale image cameraman.tif has been used in all experiments.

Watermark: A binary pattern of size MxM has been generated in the methodology itself.

Figure 7 shows the watermarked image, and extracted results with no attack.



LGAD-W Parameters used:

1. Block Size: 8x8
2. Embedding Strength: 0.15
3. Variance Percentile: 40
4. LBP percentile: 40
5. Arnold iterations: 5
6. Eligible blocks: 155
7. Watermark size: autogenerated, here 144 bits.

Attack Suite:

- JPEG Compressions: 90 to 30
- Gaussian Noise
- Salt and Pepper Noise
- Cropping

The following figure 8 shows results of different attacks. Table 2 tells different evaluation metrics. Table 3 shows corresponding results of wide varied attacks.

Evaluation Metric	**Significance**	**Formula**
PSNR (Peak Signal to Noise Ratio)	Measures pixel-wise distortion between original image I and watermarked image I_w	$PSNR = 10 \cdot \log_{10}(1 / MSE)$ *(dB)*
Normalised Correlation	Quantifies similarity between original watermark W_orig and extracted watermark W_ext	$NC = (\Sigma_i W_orig(i,j) \cdot W_ext(i,j)) / (sqrt(\Sigma_i W_orig^2) \cdot sqrt(\Sigma_i W_ext^2))$
Bit Correct Rate (BCR)	Fraction of bits correctly extracted	$BCR = (1 / (M \cdot M)) \cdot \Sigma_i [(W_orig(i,j) > 0.5) == (W_ext(i,j) > 0.5)]$

Table 2: Evaluation Metrics

fig 8. Different Attacks and Corresponding Results. Top images are attacked watermarked host images and bottom are extracted watermark

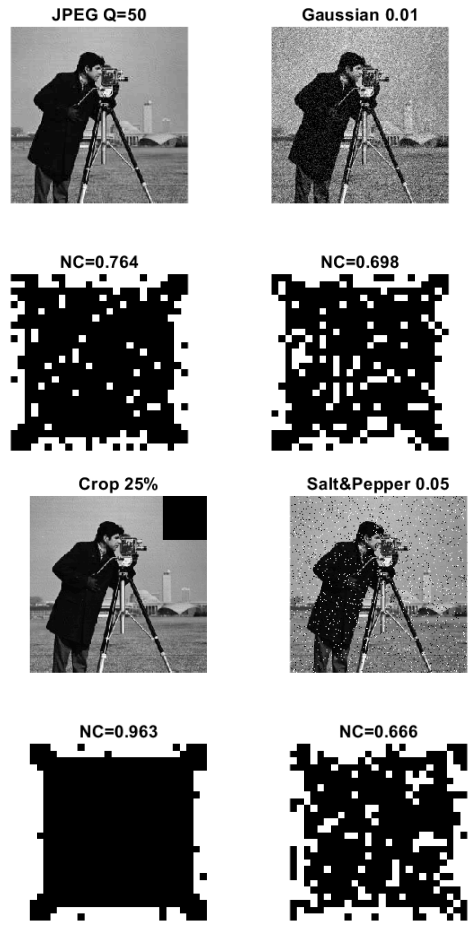


Table 3 Different Attacks and achieved Results

Attack	**NC**	**BCR**
JPEG Q=90	*1.0000*	1.0000
JPEG Q=80	*1.0000*	*1.0000*
JPEG Q=70	*1.0000*	*1.0000*
JPEG Q=60	*0.8657*	*0.9379*
JPEG Q=50	*0.7640*	*0.8891*

JPEG Q=40	*0.6279*	*0.8299*
JPEG Q=30	*0.4735*	*0.7456*
Gaussian var=0.001	*1.0000*	*1.0000*
Gaussian var=0.005	*0.8440*	*0.9216*
Gaussian var=0.010	*0.6980*	*0.8328*
Salt&Pepper d=0.01	*0.9324*	*0.9675*
Salt&Pepper d=0.02	*0.8192*	*0.9083*
Salt&Pepper d=0.05	*0.6656*	*0.8092*
Median 3x3	*0.5032*	*0.6820*
Median 5x5	*0.1678*	*0.2899*
Crop 10%	*0.9967*	*0.9985*
Crop 25%	*0.9631*	*0.9837*
Crop 50%	*0.8584*	*0.9408*
Histogram Equalization	*0.9584*	*0.9808*

Effect of Embedding Strength has been discussed in below table 4.

Table 4: Effect of embedding strength

Embedding Strength	**PSNR (dB)**	**NC (clean)**
0.05	*Upto 48.21*	*0.921*
0.10	*Upto 44.85*	*0.987*
0.15	*Upto 42.37*	*1.00*
0.20	*Upto 39.56*	*1.00*

V. DISCUSSION AND CONCLUSIONS

This paper presented LGAD-W, a blind image watermarking scheme that integrates two-factor block selection (variance + LBP complexity) with differential DCT embedding and Arnold scrambling. It has been found that high imperceptibility has been achieved, with PSNR > 42 dB with watermark invisible to human observers, along with strong robustness against common signal processing operations and malicious attacks. The extraction process is simple too, with the use of only a sign test. Thus, the proposed method is suitable for real word applications and has achieved very good results.

VI. REFERENCES

[1] N. A. Loan, N. N. Hurrah, S. A. Parah, J. W. Lee, and G. M. Bhat, "Secure and robust digital image watermarking using coefficient differencing and chaotic map," IEEE Access, vol. 6, pp. 19876–19897, 2018.

[2] C. T. Hsu and J. L. Wu, "DCT-based watermarking for video," IEEE Trans. Consumer Electron., vol. 44, no. 1, pp. 206–216, Feb. 1998.

[3] I. J. Cox, J. Kilian, T. Leighton, and T. Shamoon, "Secure spread spectrum watermarking for multimedia," IEEE Trans. Image Process., vol. 6, no. 12, pp. 1673–1687, Dec. 1997.

[4] A. Moeini, H. Saeedi, M. A. Akhaee, and R. A. Sadeghzadeh, "Robust blind image watermarking using a novel gradient-based optimizer," Multimedia Tools Appl., vol. 81, pp. 11609–11634, 2022.

[5] S. Kaur and S. Singh, "A robust digital image watermarking scheme using inter-block DCT coefficient correlation," in Proc. IEEE DELCON, 2024, pp. 1–6.

[6] M. N. Favorskaya, A. S. Zotin, and A. V. Buryachenko, "Texture analysis for digital watermarking applications," in Proc. Int. Conf. Inf. Technol. New Gener., 2014, pp. 357–362.

[7] S. Agrawal and M. Kumar, "A robust watermarking scheme based on LBP and DCT," in Proc. Int. Conf. Comput. Intell. Commun. Netw., 2016, pp. 234–239.

[8] M. H. Alkawaz, G. Sulong, and T. Saba, "A novel DCT-based watermarking scheme using LBP texture analysis," Int. J. Electr. Comput. Eng., vol. 8, no. 5, pp. 3124–3130, 2018.

[9] T. Ojala, M. Pietikäinen, and T. Mäenpää, "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns," IEEE Trans. Pattern Anal. Mach. Intell., vol. 24, no. 7, pp. 971–987, July 2002.

[10] M. Fallahpour, D. Megías, and M. Ghanbari, "Fragile watermarking using LBP for image authentication," J. Inf. Secur. Appl., vol. 40, pp. 1–10, 2018.

[11] S. Wang, B. Li, and J. Liu, "An image watermarking scheme based on LBP and DWT," J. Inf. Hiding Multimedia Signal Process., vol. 2, no. 4, pp. 363–371, 2011.

[12] B. Chen and G. W. Wornell, "Quantization index modulation: A class of provably good methods for digital watermarking and information embedding," IEEE Trans. Inf. Theory, vol. 47, no. 4, pp. 1423–1443, May 2001.

[13] M. Lebcir, A. A. K. Mahmoud, and R. Benzid, "A robust blind watermarking scheme based on DTCWT-DCT using differential embedding," Multimedia Tools Appl., vol. 77, pp. 20401–20423, 2018.

[14] H. T. Hu and L. Y. Hsu, "A DWT-based blind image watermarking scheme using differential embedding," J. Inf. Sci. Eng., vol. 32, no. 5, pp. 1179–1196, 2016.

[15] V. I. Arnold and A. Avez, Ergodic Problems of Classical Mechanics. New York: Benjamin, 1968.

[16] X. Wu, D. Sun, and L. Guo, "A chaotic watermarking scheme based on Arnold transform and DCT," J. Inf. Comput. Sci., vol. 9, no. 15, pp. 4369–4376, 2012.

[17] S. K. Sonkar and A. K. Singh, "Arnold transform and DWT based color image watermarking," in Proc. 2nd Int. Conf. Comput. Intell. Commun. Technol., 2016, pp. 234–238.